The distance between their centers $OP = d$. It is clear from the Fig. 4.57 that when the circles touch externally $OP = d = OQ + PQ = r_1 + r_2$.

5. If two circles touch internally, the distance between their centers is equal to the difference of their radii, that is $OP = r_1 - r_2$.

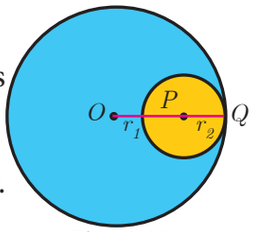


Fig. 4.58

Proof : Let two circles with centers at O and P touch each other at Q.

Let $OQ = r_1$ and $PQ = r_2$ and let $r_1 > r_2$.

The distance between their centers $OP = d$. It is clear from the Fig. 4.58 that when the circles touch internally, $OP = d = OQ - PQ$

$$OP = r_1 - r_2.$$

6. The two direct common tangents drawn to the circles are equal in length, that is $AB = CD$.

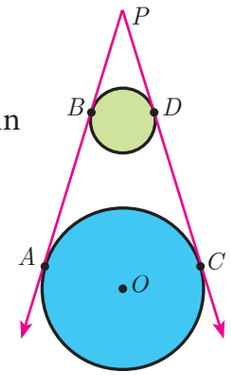


Fig. 4.59

Proof :

The lengths of tangents drawn from P to the two circles are equal.

Therefore, $PA = PC$ and $PB = PD$.

$$\Rightarrow PA - PB = PC - PD$$

$$AB = CD$$

Thinking Corner

1. Can we draw two tangents parallel to each other on a circle?
2. Can we draw two tangents perpendicular to each other on a circle?

Alternate segment

In the Fig. 4.60, the chord PQ divides the circle into two segments. The tangent AB is drawn such that it touches the circle at P.

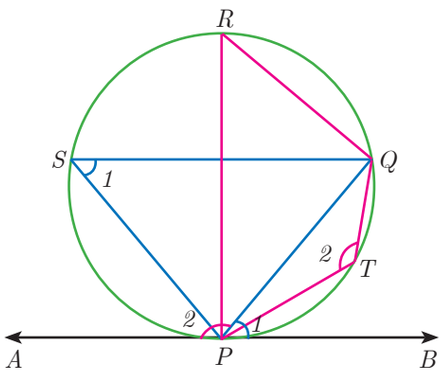


Fig. 4.60

The angle in the alternate segment for $\angle QPB$ $(\angle 1)$ is $\angle QSP$ $(\angle 1)$ and that for $\angle QPA$ $(\angle 2)$ is $\angle PTQ$ $(\angle 2)$.

Theorem 6 : Alternate Segment theorem

Statement

If a line touches a circle and from the point of contact a chord is drawn, the angles between the tangent and the chord are respectively equal to the angles in the corresponding alternate segments.

Proof

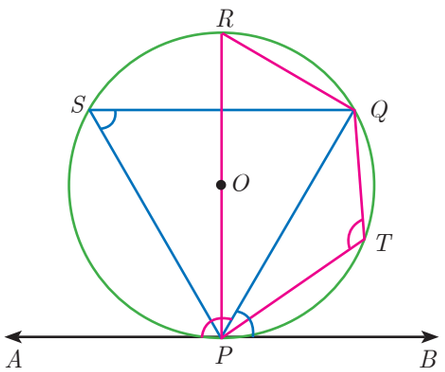


Fig. 4.61

Given : A circle with centre at O, tangent AB touches the circle at P and PQ is a chord. S and T are two points on the circle in the opposite sides of chord PQ.

4 Geometry.indd 190 16-12-2020 01:55:16 PM